

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical
Practical – XI

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Practical 11: Optimization Modeling and Simulation Modeling

Part A: Optimization Modeling

Business Problem: A company produces two products, Product A and Product B. The company wants to maximize its profit using the available machine and labour hours.

Given data:

Product	Profit per Unit	Machine Hours	Labour Hours
A	40	2	1
B	30	1	2

Available Machine Hours = 100

Available Labour Hours = 80

Practical Questions:

1. Define the decision variables for Product A and Product B.
2. Define the objective of the problem.
3. Formulate the objective function to maximize the total profit.
4. Formulate the machine-hour constraint.
5. Formulate the labour-hour constraint.
6. Add non-negative constraints for the decision variables.
7. Solve the linear programming problem using Python.
8. Find the optimal number of units of Product A and Product B.
9. Calculate the maximum profit.
10. Check whether the available machine and labour hours are satisfied.
11. Display the optimal solution in a suitable format.
12. Interpret the optimal solution in a business context.
13. Write a conclusion based on the optimization result.

Code :

```
# Practical 11: Optimization Modeling

# Part A: Optimization Modeling
from scipy.optimize import linprog
# Decision Variables
# x = Number of units of Product A
# y = Number of units of Product B
# Objective Function
# Maximize:
# Profit = 40x + 30y
#
# scipy.optimize.linprog performs minimization,
# so we use negative profit values.
profit = [-40, -30]
# Constraints
# Machine Hours:
# 2x + y <= 100
# Labour Hours:
# x + 2y <= 80
A = [
```

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        [2, 1],
        [1, 2]
    ]
    b = [100, 80]
    # Non-negative Constraints
    # x >= 0
    # y >= 0
    bounds = [
        (0, None),
        (0, None)
    ]
    # Solve Linear Programming Problem
    result = linprog(
        profit,
        A_ub=A,
        b_ub=b,
        bounds=bounds,
        method="highs"
    )
    # Display Result
    if result.success:
        product_A = result.x[0]
        product_B = result.x[1]
        maximum_profit = -result.fun
        # Calculate resource usage
        machine_used = (2 * product_A) + product_B
        labour_used = product_A + (2 * product_B)
        print("\nOPTIMIZATION MODELING")
        print("\nOptimal Production:")
        print("-----")
        print(f"Product A Units : {product_A:.0f}")
        print(f"Product B Units : {product_B:.0f}")
        print("\nProfit:")
        print("-----")
        print(f"Maximum Profit : ₹{maximum_profit:.2f}")
        print("\nResource Usage:")
        print("-----")
        print(f"Machine Hours Used : {machine_used:.0f} / 100")
        print(f"Labour Hours Used : {labour_used:.0f} / 80")
        print("\nConstraint Check:")
        print("-----")
        if machine_used <= 100:
            print("Machine Hours Constraint: Satisfied")
        else:
            print("Machine Hours Constraint: Not Satisfied")
        if labour_used <= 80:
            print("Labour Hours Constraint : Satisfied")
        else:
            print("Labour Hours Constraint : Not Satisfied")
    else:
        print("No optimal solution found.")

```

Output :

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=====
      OPTIMIZATION MODELING
=====

Optimal Production:
-----
Product A Units : 40
Product B Units : 20

Profit:
-----
Maximum Profit : ₹2200.00

Resource Usage:
-----
Machine Hours Used : 100 / 100
Labour Hours Used : 80 / 80

Constraint Check:
-----
Machine Hours Constraint: Satisfied
Labour Hours Constraint : Satisfied

```

Part B: Simulation Modeling

Business Problem: A supermarket wants to decide the appropriate inventory level for a product when daily demand is uncertain.

Practical Questions:

14. Generate random daily demand values for a product for 30 days.
15. Display the generated demand values.
16. Select an inventory level of 30 units.
17. Compare the daily demand with the available inventory.
18. Calculate the number of units sold each day.
19. Calculate the number of units left after each day.
20. Calculate the shortage when demand is greater than available inventory.
21. Calculate the total units sold and total shortage for 30 days.
22. Repeat the simulation using inventory levels of 30, 40, and 50 units.
23. Store the results of all three scenarios in a DataFrame.
24. Compare the total sales and total shortage for each inventory level.
25. Create a chart to compare the results of the three scenarios.
26. Identify how changing the inventory level affects sales and shortage.
27. Write a conclusion about the inventory level based on the simulation results.

CODE:

```
# Practical 11: Simulation Modeling
# Part B: Supermarket Inventory Simulation
import pandas as pd
import matplotlib.pyplot as plt
# 1. Load Kaggle Dataset
file_name = "retail_store_inventory.csv"
df = pd.read_csv(file_name)
print("\nDATASET INFORMATION")
print("\nFirst 5 Records:")
print(df.head())
print("\nDataset Columns:")
print(df.columns.tolist())
# 2. Generate 30 Daily Demand Values
# Select first 30 demand values from dataset
demand = df["Demand"].head(30).tolist()
days = list(range(1, 31))
print("\nDAILY DEMAND - 30 DAYS")
for day, d in zip(days, demand):
    print(f"Day {day}: {d} units")
# 3. Inventory Simulation Function
def simulate_inventory(demand, inventory_level):
    units_sold = []
    units_left = []
    shortage = []
    for daily_demand in demand:
        # Units sold cannot be greater than inventory
        sold = min(daily_demand, inventory_level)
        # Units remaining
        remaining = max(inventory_level - daily_demand, 0)
        # Shortage
        short = max(daily_demand - inventory_level, 0)
        units_sold.append(sold)
        units_left.append(remaining)
        shortage.append(short)
    return units_sold, units_left, shortage
# 4. Simulation for Inventory = 30
sales_30, left_30, shortage_30 = simulate_inventory(
    demand,
    30
)
simulation_30 = pd.DataFrame({
    "Day": days,
    "Demand": demand,
    "Inventory": 30,
    "Units Sold": sales_30,
    "Units Left": left_30,
    "Shortage": shortage_30
})
print("\nSIMULATION - INVENTORY = 30")
print(simulation_30)
print("\nTotal Units Sold:",
      simulation_30["Units Sold"].sum())
```

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print("Total Shortage:",
      simulation_30["Shortage"].sum())
# 5. Simulate Inventory Levels
# 30, 40 and 50
inventory_levels = [30, 40, 50]
results = []
for inventory in inventory_levels:
    sales, left, shortage = simulate_inventory(
        demand,
        inventory
    )
    total_sales = sum(sales)
    total_shortage = sum(shortage)

    results.append({
        "Inventory Level": inventory,
        "Total Sales": total_sales,
        "Total Shortage": total_shortage
    })
# 6. Store Results in DataFrame
results_df = pd.DataFrame(results)
print("\nINVENTORY COMPARISON")
print(results_df)
# 7. Display Best Sales and Shortage Results
print("\nANALYSIS")
highest_sales = results_df.loc[
    results_df["Total Sales"].idxmax()
]
lowest_shortage = results_df.loc[
    results_df["Total Shortage"].idxmin()
]
print(
    f"Highest Total Sales: "
    f"Inventory {highest_sales['Inventory Level']} units"
)
print(
    f"Lowest Total Shortage: "
    f"Inventory {lowest_shortage['Inventory Level']} units"
)

# 8. Create Chart
plt.figure(figsize=(9, 6))
plt.plot(
    results_df["Inventory Level"],
    results_df["Total Sales"],
    marker="o",
    label="Total Sales"
)
plt.plot(
    results_df["Inventory Level"],
    results_df["Total Shortage"],
    marker="o",
    label="Total Shortage"
)
plt.xlabel("Inventory Level")
plt.ylabel("Units")

plt.title(
    "Inventory Level vs Total Sales and Total Shortage"
)
plt.legend()
plt.grid(True)
plt.show()

# 9. Save Results
results_df.to_csv(
    "inventory_simulation_results.csv",
    index=False
)

```

Output :

 DATASET INFORMATION

First 5 Records:

	Date	Store ID	Product ID	Category	Region	Inventory Level \
0	2022-01-01	5001	P0001	Electronics	North	195
1	2022-01-01	5001	P0002	Clothing	North	117
2	2022-01-01	5001	P0003	Clothing	North	247
3	2022-01-01	5001	P0004	Electronics	North	139
4	2022-01-01	5001	P0005	Groceries	North	152

	Units Sold	Units Ordered	Price	Discount	Weather Condition	Promotion \
0	102	252	72.72	5	Snowy	0
1	117	249	80.16	15	Snowy	1
2	114	612	62.94	10	Snowy	1
3	45	102	87.63	10	Snowy	0
4	65	271	54.41	0	Snowy	0

	Competitor Pricing	Seasonality	Epidemic	Demand
0	85.73	Winter	0	115
1	92.02	Winter	0	229
2	60.08	Winter	0	157
3	85.19	Winter	0	52

 ANALYSIS

Highest Total Sales: Inventory 50 units

Lowest Total Shortage: Inventory 50 units

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